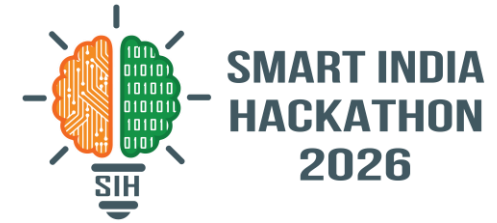


SMART INDIA HACKATHON 2026



SUKOBIN

Carriers as the sensor network

Problem Statement ID – SIH26-MDoNER-01

Problem Statement Title – AI-Enabled Logistics Accessibility Intelligence Platform for the North Eastern Region

Theme – Transportation & Logistics

PS Category – Software

Team ID – <team id>

Team Name – Sukobin



EVERY VEHICLE ON THE ROAD IS A SENSOR

Sukobin



THE GAP

The NER has no road-condition sensor network. A district usually learns a road is shut when a truck is already stuck on it, and instrumenting 3,500 km of hill highway is not affordable.

THE IDEA

Nobody drives for us. Anyone already travelling A to B enters their vehicle and route, and sees only the parcels whose pickup and drop lie along that road, up to what the vehicle holds.

THE TURN THAT MAKES IT INTELLIGENCE

Those carriers stream GPS while they drive. The rolling median of their speed against each road's baseline is a live accessibility reading. One stream, two products: the goods move, and the road gets measured.

WHY THIS IS DIFFERENT

- **Coverage without hardware** — no roadside sensors, no dedicated fleet.
- **Cost per reading falls to zero** — the driver was making the trip anyway.
- **It improves as it is used** — more carriers means denser sensing.

HOW ONE TRIP BECOMES BOTH



The carrier network and the sensor network are the same network. That is the whole idea.

BUILT WITH

Backend Node.js · Express 5 · MongoDB with 2dsphere · JWT

Web React · Vite · MapLibre GL

Mobile Kotlin + XML — four apps: customer, merchant, driver, officer

Cloud Render · MongoDB Atlas · Cloudinary · FCM push

THE MODEL

Type Logistic regression, benchmarked against gradient-boosted stumps

Features 18 — antecedent rain, burst intensity, slope, terrain, closure history

Trained on 109,116 road-days of observed weather, 42 stretches x 877 days

Scored AUC 0.883 · Brier 0.092 · held out after 2026-03-01

REAL DATA, NOT MOCK

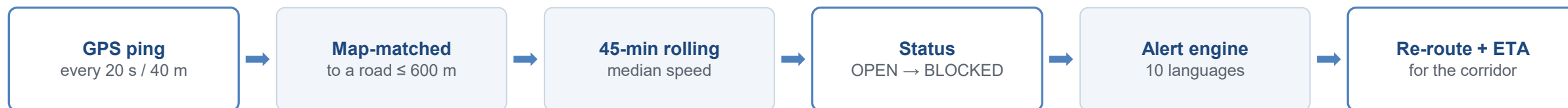
Weather Open-Meteo archive + forecast, hourly

Roads OSRM geometry — 42 stretches / 3,567 km

Vehicles VAHAN registration lookup

Field Photos, GPS and voice from officers and drivers

FROM ONE GPS PING TO A ROAD STATUS, AN ALERT AND A ROUTE



Trust rule — a reading needs at least 4 samples from at least 2 distinct vehicles, so one parked driver can never close a highway. A driver's hazard report is capped at RESTRICTED until an officer confirms it.

FEASIBLE BECAUSE IT IS ALREADY BUILT AND MEASURED

42 stretches · 3,567 km real OSRM geometry across 12 corridors, 82 districts, 8 states

4 Android apps + dashboard customer, merchant, driver, officer, control room

195 / 195 backend tests passing across 10 suites

10 languages · 8,424 units every screen, zero English fallbacks

100 % coverage status known, live vehicle data and forecast on every road

SCALING COSTS ALMOST NOTHING

- No hardware to install, maintain or power.
- A new district needs its road geometry seeded — hours, not procurement.
- Sensing density rises with adoption, at no extra cost.

RISKS, AND WHAT ANSWERS EACH ONE

No public register of past road closures

Labels drawn from a rainfall-threshold hazard function; verified field reports override them. Stated openly on the dashboard.

One parked driver could read as a closure

A status needs ≥ 4 samples from ≥ 2 vehicles, else it is discarded.

Hill districts drop off the network

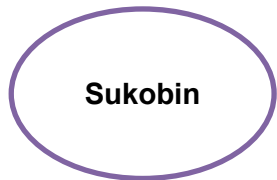
Reports queue on the phone with a client ID and sync later; duplicates are rejected, so nothing is filed twice.

Officers and drivers do not share a language

Ten languages including Khasi, Mizo, Nagamese and Kokborok — spoken aloud as well as written.

A quiet road has no vehicles to sense it

The weather model forecasts every road regardless of traffic; the dashboard shows honestly what is sensed and what is not.



IMPACT AND BENEFITS



WHO IT REACHES

District administration

A lifeline corridor's closure risk 72 hours out, ranked by how many people lose their only road.

Field officers

One screen: weak points ranked with the reason each one scored, and a queue of reports to confirm.

Drivers and small operators

Income from a journey already being made — no fleet, no contract, no empty return leg.

People in remote blocks

Medicines, rations and produce arrive, and an alert in the language spoken at home when they will not.

BENEFITS

Social

8 states · 82 districts. Alerts and voice reporting in 10 languages, including four with no other software support.

Economic

No dedicated fleet, so the marginal cost of moving one more parcel is near zero. Spare capacity in vehicles already moving becomes income.

Environmental

Fewer dedicated trips: a parcel rides in a vehicle that was making the journey regardless, instead of a second one setting out.

Governance

Every incident carries a photo, GPS fix, timestamp and the officer who confirmed it — an evidence trail, not a phone call.

DATA SOURCES THE PLATFORM RUNS ON

Open-Meteo Historical Weather API

open-meteo.com/en/docs/historical-weather-api — observed hourly rainfall, snowfall and temperature; the model's training set

OSRM — Open Source Routing Machine

project-osrm.org — road geometry and alternates for all 42 stretches

VAHAN / Parivahan registration lookup

parivahan.gov.in — vehicle class and capacity at driver sign-up

MapLibre GL JS + Esri Dark Gray Canvas

maplibre.org — the GIS layer of the control dashboard

PROBLEM AND DOMAIN RESEARCH

MDoNER — problem statement and regional context

mdoner.gov.in — NER connectivity gaps and infrastructure priorities

NHIDCL / MoRTH national highway network

nhidcl.com · morth.nic.in — corridor alignments and NH numbering used to seed the network

IMD rainfall climatology for the North East

mausam.imd.gov.in — monsoon windows and the rain thresholds behind the hazard function

GSI landslide susceptibility mapping

gsi.gov.in — which stretches are treated as landslide-prone

Working prototype — four Android apps, a control dashboard and the backend are built and running. The forecast model, the probe-sensing engine and the corridor matcher are covered by 195 automated tests.